

ASHE COUNTY AP, NC, USA

WMO: 723146

Lat: 36.432N

Lon: 81.419W

Elev: 3180

StdP: 13.08

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53892

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	15.5	-8.5	3.9	25.3	-2.6	5.4	24.2	37.3	29.5	34.0	29.9	11.0	320	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	82.4	67.2	81.2	67.2	79.3	66.7	71.4	77.8	70.4	77.0	69.3	76.0	6.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.0	124.4	75.2	68.3	117.0	73.7	66.9	111.6	72.6	37.6	78.4	36.5	77.1	35.5	75.7	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	23.6	19.0	5.9	86.2	6.2	3.7	1.4	88.8	-2.3	91.0	-5.8	93.0	-10.3	95.7
			4.0	73.1	5.5	1.9	0.1	74.5	-3.2	75.6	-6.2	76.7	-10.2	78.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	33.9	37.4	44.0	53.5	61.4	68.0	70.3	69.0	63.7	54.2	44.9	38.2
	DBStd	15.01	10.90	10.01	9.95	8.80	6.67	4.87	4.33	4.25	5.70	7.86	9.10	9.60
	HDD50	1811	506	364	237	64	7	0	0	0	1	49	201	382
	HDD65	4792	964	771	651	350	145	25	9	14	89	340	602	832
	CDD50	3012	8	12	51	170	359	539	629	588	411	180	50	15
	CDD65	520	0	0	0	6	33	115	173	137	49	7	0	0
	CDH74	2453	0	0	1	27	137	582	840	632	210	23	1	0
	CDH80	302	0	0	0	0	7	90	115	77	12	1	0	0
Wind	WSAvg	5.9	8.8	8.3	7.7	7.1	5.4	4.5	4.0	3.0	3.1	5.3	6.2	7.2
Precipitation	PrecAvg	50.2	4.1	3.1	5.0	3.8	4.2	5.0	4.8	4.7	4.3	3.4	3.7	4.0
	PrecMax	64.8	10.9	7.2	9.1	7.9	9.6	9.6	7.4	11.1	16.1	13.0	9.9	8.4
	PrecMin	37.2	1.0	1.0	1.3	0.6	1.4	1.3	2.6	0.8	0.4	0.0	0.5	1.3
	PrecStd	8.0	2.6	1.5	1.9	1.9	2.1	2.3	1.4	2.6	3.3	2.8	2.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	64.3	71.9	77.2	81.0	86.2	86.3	84.4	81.7	77.3	70.4	64.0
		MCWB	52.4	53.8	55.6	57.7	63.0	67.1	68.4	69.4	65.3	64.8	56.7	53.8
	2%	DB	56.8	59.6	66.1	73.3	77.5	82.1	82.3	81.8	79.1	72.7	64.5	59.6
		MCWB	49.9	52.0	52.6	57.4	62.3	66.2	68.0	68.3	65.2	60.9	53.9	52.9
	5%	DB	54.2	55.8	62.8	71.5	75.0	80.5	81.1	80.6	76.6	69.4	61.7	55.4
		MCWB	48.3	48.8	51.0	56.4	61.6	65.7	68.2	68.3	64.7	58.5	52.7	49.4
	10%	DB	49.9	52.5	58.8	67.5	72.6	77.3	79.1	78.4	73.3	65.8	58.6	53.4
		MCWB	44.5	45.6	48.6	55.4	60.9	65.4	67.7	67.7	64.3	58.6	51.1	48.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	57.5	57.7	62.7	68.2	71.5	72.9	73.1	71.6	68.7	61.7	59.1
		MCDB	59.1	61.1	66.8	69.8	76.2	78.6	79.4	79.8	75.7	73.5	64.5	61.4
	2%	WB	52.6	54.4	55.4	60.4	66.2	69.7	71.3	71.5	68.9	65.6	58.3	55.0
		MCDB	56.4	57.9	62.1	68.6	73.1	77.4	78.0	77.8	74.0	68.8	62.5	58.1
	5%	WB	49.2	50.6	53.5	58.6	64.4	68.4	70.3	70.3	67.1	62.8	55.7	51.8
		MCDB	52.4	54.8	58.9	66.6	71.0	75.9	77.1	76.8	72.2	66.1	59.0	54.3
	10%	WB	45.1	46.8	51.3	56.8	62.9	67.2	69.2	69.1	65.9	60.0	53.0	48.2
		MCDB	47.9	51.4	57.3	64.4	69.6	74.5	76.0	75.5	70.9	64.3	56.9	52.0
Mean Daily Temperature Range	5% DB	MDBR	15.0	15.7	16.2	17.1	16.5	15.8	14.7	14.8	15.5	15.8	16.6	14.3
		MCDBR	17.9	19.2	19.4	20.0	18.8	18.4	16.6	17.5	19.1	19.2	19.3	17.4
		MCWBR	13.3	13.6	11.4	9.9	8.8	7.9	7.1	8.0	8.8	10.0	12.0	12.8
	5% WB	MCDBR	16.3	18.0	16.5	17.0	16.5	16.0	14.8	15.0	15.9	14.5	15.0	15.7
		MCWBR	14.5	15.3	12.0	10.3	9.1	8.2	7.5	7.9	9.0	9.7	11.9	14.2
Clear-Sky Solar Irradiance	taub	0.278	0.284	0.308	0.352	0.389	0.419	0.451	0.450	0.358	0.320	0.288	0.274	
	taud	2.557	2.532	2.506	2.384	2.282	2.223	2.169	2.147	2.401	2.511	2.575	2.597	
	Ebn at Noon	296	306	306	295	284	274	264	262	284	288	289	291	
	Edh at Noon	24	28	31	37	42	45	47	47	35	28	23	22	
All-Sky Solar Radiation	RadAvg	753	953	1304	1631	1784	1916	1781	1656	1419	1141	862	652	
	RadStd	56	108	79	129	135	134	129	84	131	137	78	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+270	+164

Nomenclature: See separate page